Vivek Kumar Singh
Enterprise Architect | Cloud & Data Engineering Leader | Big Data | Spark | Azure | Databricks | Generative AI
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Executive Summary

Strategic and hands-on **Enterprise Architect** with **18+ years** of experience designing, delivering, and scaling complex **data, cloud, and AI platforms**. Proven expertise in **multi-cloud transformation** (AWS, Azure, GCP), **Big Data architecture** (Cloudera, Spark, Hive, Delta Lake), and **enterprise-scale AI/GenAI integrations**.

Specializes in **modernizing legacy ecosystems** to **cloud-native, secure, and high-performance solutions**. Skilled in **leading data lakehouse implementations**, orchestrating **Spark-based ETL/ELT pipelines**, and managing **cross-functional teams in Agile programs**. Recent focus includes implementing **Generative AI, RAG, LangChain, LlamaIndex, AgentAI, VectorDB**, and **prompt engineering** for enterprise decision-making.

Key Strengths & Expertise

Enterprise Architecture & Strategy

- Led architectural design for **large-scale enterprise platforms** (APIs, microservices, SOA).
- Defined **technical roadmaps**, ensuring **scalability, security, and business alignment**.

Multi-Cloud & Big Data Leadership

- Extensive experience in **AWS, Azure, GCP, Cloudera CDH (5.9.X, 6.X), Spark, Scala, PySpark, Kafka, Hadoop, HDFS, Hive, Impala, MongoDB, HBASE**.
- Cloud-native services:
    - **AWS:** S3, Glue, Athena, Lambda, Redshift, EMR, Databricks Delta Lake
    - **Azure:** Data Lake, Synapse, Databricks – Unity Catalog, Jobs, Notebook, Lakeflow
    - **GCP:** BigQuery, Dataflow, Dataproc
- Expertise in **Linux, Jenkins, Git/GitHub, Airflow, Control-M, Agile/Scrum (JIRA), Cycode, performance tuning**

Data Engineering & AI/GenAI

- Designed **ETL/ELT pipelines**, **data lake/lakehouse solutions**, and orchestrated workflows using **Talend Big Data, Apache Airflow, Control-M**.

- Implemented **Generative AI solutions** using **LLMs, RAG, LangChain, LlamaIndex, AgentAI, VectorDB, prompt engineering/tuning**.
- Prepared curated datasets for ML models and integrated analytics outputs.

Full SDLC & Agile Excellence

- Led teams across **all SDLC phases**, from requirement analysis to production support.
- Certified in **Agile Scrum Methodology**, delivering solutions in iterative sprints.

Leadership & Communication

- Strong **presentation, problem-solving, and analytical skills**.
- Proven ability to **lead teams, capture requirements, and communicate effectively** across all organizational levels.

Technical Skills

Big Data & Cloud: Hadoop, Spark, Scala, PySpark, Python, Hive, HDFS, Impala, Cloudera CDH, HBASE, Kafka, Delta Lake, Databricks
AI / GenAI: LangChain, LlamaIndex, Hugging Face, OpenAI GPT, Azure OpenAI, RAG, AgentAI, VectorDB, Prompt Engineering & Tuning
ETL & Orchestration: Talend Big Data, Apache Airflow, Control-M, Sqoop, Apache NiFi
Databases: Oracle, SQL Server, PostgreSQL, MongoDB, Hive, Impala, Snowflake, Cassandra
Programming & Frameworks: Python, Scala, Java, JavaScript, Spring Boot, Spring MVC, Hibernate
DevOps & Tools: Jenkins, Bamboo, Git, Bitbucket, Sonar, Eclipse, IntelliJ
Web & APIs: RESTful/SOAP Web Services, WSO2, JSP, Servlets
Visualization: Power BI, Qlik Sense, Tableau

Professional Experience

Project: Retail Pricing Analytics Platform – NielsenIQ

Client: NielsenIQ (US Retail Analytics Domain)

Role: Solution Architect

Duration: Jan 2025 - Present.

Project Overview:

The project focused on building a Retail Pricing Analytics Platform for major US retailers such as Target, Walmart, and other nationwide retail chains. The goal was to ingest, process, and analyze large-scale product pricing and sales data from multiple retail sources to generate competitive pricing intelligence, trend insights, and product comparison analytics for various FMCG categories. The solution enabled NielsenIQ to track price variations, promotions, and product performance across multiple retailers, helping clients make data-driven pricing and marketing decisions.

Key Responsibilities:

- Designed and architected an end-to-end data analytics platform using Azure and Databricks for large-scale retail data processing.
- Defined data ingestion, transformation, and analytics pipelines leveraging Databricks medallion architecture.
- Implemented data modeling (Bronze → Silver → Gold layers) to ensure data quality, reusability, and traceability.
- Collaborated with business and data science teams to enable comparative pricing analysis and trend forecasting models.
- Integrated GitHub and Cycode for secure CI/CD and code governance.
- Managed sprint deliverables and architecture documentation using Jira and Confluence.
- Oversaw environment setup and automation on Linux-based Azure infrastructure.

Technology Stack:

- Microsoft Azure – Data Lake Storage Gen2, Key Vault, DevOps/GitHub Actions
- Databricks – Workspace, Delta Lake, Notebooks, Jobs, Repos, Clusters, SQL, Unity Catalog
- PySpark / Spark SQL – Data ingestion, cleansing, and transformation
- Cycode – Source code security and compliance enforcement
- Linux Shell Scripting – Automation and orchestration
- Project Management – Jira, Confluence

TCS | Enterprise Architect
May 2022 – Dec 2024
Project: TCS Connected Intelligence Platform™

Project Description:
TCS Connected Intelligence Platform™ is a Unified Data Lake and Analytics Platform enabling rapid development of enterprise data and analytic use cases through a low-code approach. It unifies diverse big data technologies into a simplified end-to-end data management and analytics environment, turning raw data into actionable insights for continuous business innovation. The platform incorporates enterprise AI assistants powered by large language models (LLMs), enabling natural language interaction with structured and unstructured data. A retrieval-augmented (RAG) architecture with semantic search and vector-based indexing allows contextual question answering, intelligent document retrieval, and conversational analytics across multiple data sources.

Roles & Responsibilities:

- Led architectural reviews, designed solution architectures, and provided strategic technology recommendations.
- Managed and optimized platform performance for efficiency and scalability.
- Developed PoC solutions to demonstrate technical feasibility and innovation.
- Ensured alignment with enterprise standards and drove technology adoption and best practices.

Environment: GenAI, LLMs, LangChain, LlamaIndex, Hugging Face, OpenAI GPT, Azure OpenAI, pgvector, Databricks (Unity Catalog, Jobs, Notebook, Delta, Lakeflow), Delta Lake, Spark, PySpark, Kafka, PostgreSQL, Hive, MongoDB, Azure, AWS, GCP, Python, Scala

Legato Health Technology (Anthem India) | Tech Lead / Sr. Big Data Engineer
Jan 2021 – Apr 2022
Project: CQADM (Clinical Quality Analytical Data Mart)
Project Description:
Developed and led multiple Spark Scala/Python ETL pipelines to load data from S3 bucket, Redshift, RDBMS, Kafka, MongoDB into Hive tables.

Roles & Responsibilities:

- Optimized Spark jobs and configured automated production runs using Airflow and Control-M.
- Managed large-scale healthcare data (MCAID, MCARE, commercial health, EMR, LAB) for BI and ML applications.
- Developed Spring Boot REST services for reporting.
- Collaborated in Agile environment for sprint planning, task delivery, and issue resolution.

Environment: Spark 2.4, Scala, PySpark, Python, HDFS, Hive, Impala, Cloudera CDH 6.x, Bamboo, Bitbucket, Apache Airflow, Control-M, AWS CLI

Anthem Inc. (USA) | Sr. Big Data Engineer (Permanent)
Jun 2018 – Dec 2020
Project: HEDIS Supplemental
Project Description:

Built Spark ETL pipelines for Oracle/CSV data to HDFS and Hive to support clinical quality analytics and HEDIS runs.

Roles & Responsibilities:

- Automated jobs using CONTROL-M and CI/CD with Bamboo & Bitbucket.
- Developed Spring Boot REST services for reporting.
- Participated in Agile: sprint planning, JIRA, SIT/UAT support.

Environment: Spark (Scala/SQL), HDFS, Hive, SQL Server, CONTROL-M, Spring Boot, Bamboo, Bitbucket, Linux

First Data Corporation, Atlanta, Georgia | Senior Big Data Engineer(Cognizant)
May 2016 – Jun 2018
Project: DATAHUB
Project Description:
DATAHUB is a big data storage system containing live and historical data in Hadoop. Data ingestion was implemented via Talend Big Data (batch and stream) and Kafka for real-time processing. Mainframe files up to 40 GB were parsed and loaded to HDFS as Parquet for fraud detection and reporting.

Roles & Responsibilities:

- Developed Talend ingestion jobs with Hive, Impala, HDFS, and dynamic schemas.
- Processed real-time data from Kafka using Spark Streaming.
- Managed Cloudera CDH cluster, HDFS partitions, Sentry/ACL security, job monitoring, and TAC configuration.
- CI/CD with Git/Nexus and deployed jobs to TAC.

Environment: Java 1.8, Hadoop (Hive, Impala, HDFS), Altiscale Hadoop Cloud, Kafka, Spark, Flume, Sqoop, Linux, Python, Talend Big Data, HP Voltage API

State Street Corporation, Boston, MA | Senior Developer(Cognizant)
May 2015 – Apr 2016
Project: LCCS (Local Custody Control System)
Project Description:
Handles local market processing and automation while insulating core custody systems. Interfaces with GSMAC and MTEX for Canada, UK, and Germany markets.

Roles & Responsibilities:

- Developed Java/J2EE code, Cloud API, GWT, IBM MQ, Oracle stored procedures.

- Deployed via Jenkins and SVN; post-production support and testing.

Environment: Java 1.6, GWT 2.4, IBM MQ, JMS, Oracle, Cloud, VeraCode, Junit, Sonar, Jenkins

Barclays Capital, India | Lead Developer(Cognizant)
Jul 2012 – Mar 2015
Project: EDOC
Project Description:
Web client for IBM CMOD enterprise content management; users can upload/download documents with SSO authentication.

Roles & Responsibilities:

- Developed enhancements and bug fixes using Java/J2EE, Spring, Spring Security, ODWEK API.
- Deployed to Apache Tomcat, wrote JUnit tests, SVN source control, coordinated offshore team.

Environment: Java 1.6, Spring 3.1, Maven, GWT 2.3, UNIX, ODWEK 8.5, Adobe Live, TeamCity, JUnit

Axis Bank, India | Senior Developer(Cognizant)
Nov 2011 – Jun 2012
Project: LOS (Loan Organization System)
Project Description:
Manages customer loan proposals across different business lines with audit and reporting.

Roles & Responsibilities:

- Developed BPM using Cordys, designed database in MS SQL Server, created BPS process for forms.
- Participated in Scrum meetings, provided production support, unit testing.

Environment: Java, Cordys BPM, JDBC, MS SQL Server 2008, XML

PERSYSTENT SYSTEMS | Team Lead
Jan 2010 – Oct 2011
Project: Prefilled FD Form Application
Project Description:
Developed application using Java, JSP, Struts, Hibernate. Designed database schema and Oracle 10g stored procedures.

Arion Systems Pvt. Ltd. | Senior Software Engineer
May 2010 – Jan 2011
Project: Client Service Portal (CSP)
Project Description:
Developed application for MIS reporting using Java, JSP, JSF. Authored stored procedures, triggers, and views in Oracle 10g.

India Games Ltd. | Software Developer
Nov 2006 – Apr 2010
Project: Mobile Games Development
Project Description:
Developed and ported mobile games on J2ME.

Education

Master of Computer Application – Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal (M.P), India